

# testMayo2016-sol.pdf



Anónimo



Bases de Datos



2º Grado en Ingeniería Informática del Software



Escuela de Ingeniería Informática  
Universidad de Oviedo



MÁSTER EN

## Inteligencia Artificial & Data Management

MADRID

Formamos  
**talento** para un futuro  
**Sostenible**

saber más



# PEREZA: la excusa perfecta para no ir a clase

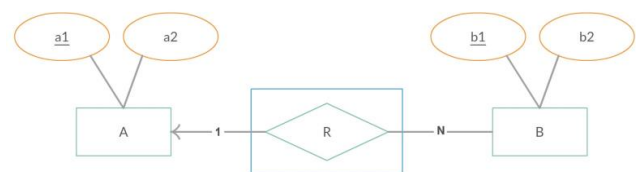
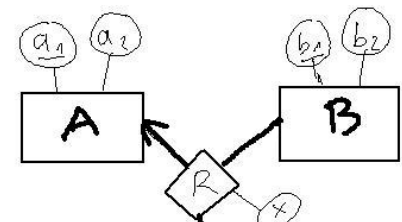
Escuela de Ingeniería Informática de Oviedo

Year 2015-2016

Databases

THEORY - TEST – Model X

- 1) The metadata stored in the data dictionary of a DBMS
  - a) contains information describing the conceptual schema of the database
- 2) Given a relational schema  $R$ , and a decomposition of  $R$  into  $R_1$  and  $R_2$ , and given any  $r(R)$ ,  $r_1(R_1)$  and  $r_2(R_2)$ , a decomposition is a LOSSY JOIN (the opposite of lossless) decomposition due to
  - a)  $r_1 \bowtie r_2$  (natural join) has MORE tuples than the original  $r$
- 3) Given a referential integrity with a referenced relation  $r_1$  and primary key  $K$ , and a referencing relation  $r_2$  with foreign key  $\alpha$ 
  - a) None of the others is right
- 4) Given the relational schema  $R = \{ A, B, C, D, E \}$ , and any possible  $r(R)$  relation
  - a)  $\{ A, B, C, D, E \}$  is a superkey
- 5) A relation that is in 3NF
  - a) None of the others is right
- 6) The exclusion constraint  $\oplus$  in the E-R model
  - a) Can be used together with the maximum cardinality constraint
- 7) Given  $R = (A, B, C, D, E)$  and a set of dependencies  $F = \{ AB \twoheadrightarrow DE, E \twoheadrightarrow B \}$ 
  - a)  $R$  is not in BCNF nor in 3NF
- 8) Given the following Entity-Relationship diagram, in the usual mapping into tables mechanisms, a valid mapping is ( $x$  attribute is NOT part of the key for  $R$ ):
  - a)  $A(\underline{a1}, a2)$   $B(\underline{b1}, b2)$   $C(\underline{c1}, c2)$   $R(a1, b1, c1, x)$   $R$  key:  $b1+c1$
- 9) Given this Entity-Relationship ( $A, B$  – aggregated  $R$  relationship), a valid mapping into tables of the diagram is:
  - a)  $A(\underline{a1}, a2)$   $B(\underline{b1}, b2, a1)$
- 10) Given  $F = \{ A B \twoheadrightarrow D, B C \twoheadrightarrow A, A C \twoheadrightarrow B, C \twoheadrightarrow D \}$ , a canonical (minimal) cover of this set of functional dependencies is
  - a)  $F$  is already a canonical (minimal) cover



Doble carne,  
cheddar  
americano,  
taquitos de  
bacon y salsa  
de bacon  
ahumado.

La excusa  
perfecta  
para que  
vengas a  
probarla.

Si te da la Pereza, que sea  
en forma de hamburguesa.  
**Pecar es divino.**

